

5. ALU:

→ Functional features: 8 bit data, carry, overflow, zero flags
ADD, SUB, AND, OR, XOR, NAND, LSL, LSR

→ Scoreboard strategy:

→ Collect transactions of A, B, and op code.

→ Calculate golden reference.

→ Compare DUT output with golden reference.

→ Directed tests:

→ ADD $\Rightarrow FF + 1$

→ SUB $\Rightarrow 00 - 1$

→ AND $\Rightarrow 55 \& AA$

→ OR & XOR $\Rightarrow 55 \mid AA, 55 \wedge AA$

→ XLL $\Rightarrow A=00, B=00$

$A=FF, B=FF$

→ Coverage:

→ bin A

→ bin B

→ bin op-code

→ bin Result

→ Cross A X B X op-code